

JEE MAIN REVISION MASTER: LIMITS, CONTINUITY AND DIFFERENTIABILITY

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 271 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Continuity & Differentiability at a Point	158 PYQs	Continuity & Differentiability
Very High	Evaluation of Limits & L'Hopital's Rule	116 PYQs	Limits

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Continuity & Differentiability at a Point	$\text{Continuous at } a \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a), \quad \text{Differentiability} \implies \text{Continuity}$
Evaluation of Limits & L'Hopital's Rule	$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1, \quad \lim_{x \rightarrow a} f(x)^{g(x)} = e^{\lim_{x \rightarrow a} (f(x) - 1)g(x)} \quad (\text{for } 1^\infty)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Continuity & Differentiability at a Point (Very High Yield — 158 PYQs)

Core Principle: Epsilon-delta and limit criteria for continuity, left/right hand derivatives, and graphical identification of non-differentiable sharp points.

■ **Caution / JEE Pitfall:** Differentiability implies continuity, but continuity DOES NOT imply differentiability (e.g. $f(x) = |x|$ is continuous at $x=0$ but not differentiable).

■ **Speed Trick:** The greatest integer function $[x]$ is discontinuous at all integer points $x \in \mathbb{Z}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "For the function $f(x)=e^{\sin|x|-|x|}, x \in R$, consider the following statements : Statement I : f is differentiable for all $x \in R$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let $y=y(x)$ be a differentiable function in the interval $(0,\infty)$ such that $y(1)=2$, and $\lim_{t \rightarrow x} \frac{y(t^2)-y(x^2)}{t^2-x^2} = \frac{1}{2}$..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "If $f(x)=\frac{|x|+x^2-2(\sin|x|)(\cos|x|)}{x}, x \neq 0$, $x=0$ is continuous at $x=0$, then $a+b$ i..." [Ans: N/A]

3.2 Evaluation of Limits & L'Hopital's Rule (Very High Yield — 116 PYQs)

Core Principle: Resolution of indeterminate forms ($0/0$, ∞/∞ , 1^∞), standard trigonometric and exponential limits, and Series expansion.

■ **Caution / JEE Pitfall:** Before applying L'Hopital's rule, ALWAYS check that the limit is in an indeterminate form ($0/0$ or ∞/∞). Differentiating a determinate form produces an incorrect answer.

■ **Speed Trick:** Maclaurin expansions are often faster than repeated L'Hopital: $\sin x = x - (x^3/6)$, $\cos x = 1 - (x^2/2)$, $e^x = 1 + x + (x^2/2)$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "The product of all possible values of α , for which $\lim_{x \rightarrow 0} \frac{1 - \cos(\alpha x) \cos((\alpha+1)x) \cos(\alpha x)}{x^3}$ is finite, is $\dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "The value of $\lim_{x \rightarrow 0} \frac{x^2 \sin^2 x - \sin^2 2x}{x^2}$ is $\dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "If $\lim_{x \rightarrow 2} \frac{\sin(x^3 - 5x^2 + ax + b)}{(\sqrt{x-1} - 1) \log(e)(x-1)} = m$, then $a+b+m$ is equal to $\dots$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

Let $f(x) = x^3 + 8$; $x < 0$, $x^2 - 4$; $x \geq 0$, and $g(x) = (x-8)^{1/3}$; $x < 0$, $(x+4)^{1/2}$; $x \geq 0$. Then the number of points, where the function $g \circ f$ is discontinuous, is _____.

Correct Answer: 3

Solution Analysis: We need to find the points where the composite function $g \circ f$ is discontinuous. That means we first find $(g \circ f)(x) = g(f(x))$. So we must see, for each value of x , which branch of g will be used. 1. Given functions $f(x) = x^3 + 8$, $x < 0$, $x^2 - 4$, $x \geq 0$ and $g(x) = (x-8)^{1/3}$, $x < 0$, $(x+4)^{1/2}$, $x \geq 0$. Now, $(g \circ f)(x) = g(f(x))$. The definition of g depends on whether its input is negative or non-negative. So we must check the sign of $f(x)$. 2. Find where $f(x) < 0$ and where $f(x) \geq 0$
Case 1: $x < 0$

Benchmark Problem 2: JEE Main 2025 (Online) 3rd April Evening Shift (Q#1)

If $\lim_{x \rightarrow 0} \frac{\tan x}{x} \leq p$, then $96 \log_e p$ is equal to _____

Correct Answer: 32

Solution Analysis: To solve the given limit problem, we start by analyzing the expression: $\lim_{x \rightarrow 0} \frac{\tan x}{x} = p$. This limit exhibits the indeterminate form 1^∞ . To handle this form, we use the transformation: $p = e^{\lim_{x \rightarrow 0} ((\tan x / x) - 1) \frac{1}{x^2}}$. Expanding $\tan x$ using its Taylor series near $x=0$, we have: $\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots$. Substituting this expansion into the limit, we get: $\frac{\tan x - x}{x^3} = \frac{(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots) - x}{x^3} = \frac{\frac{x^3}{3} + \frac{2}{15}x^5 + \dots}{x^3} = \frac{1}{3} + \frac{2}{15}x^2 + \dots$